Senin, 11 MEI 2026

ibimbing

STUDENT BAG

Data exploration, Data cleaning, Data manipulation

Presented by: Yumita

POIN - POIN PENTING

1. SELF AND PROJEK OVERVIEW
2. PROBLEM STATEMENT
3. DATA UNDERSTANDING
4. DATA EXPLORATION
5. DATA CLEANING
6. DATA MANIPULATION
7. OUTPUT FINAL
8. CONCLUSION

TENTANG SAYA

YUMITA

PENDIDIKAN

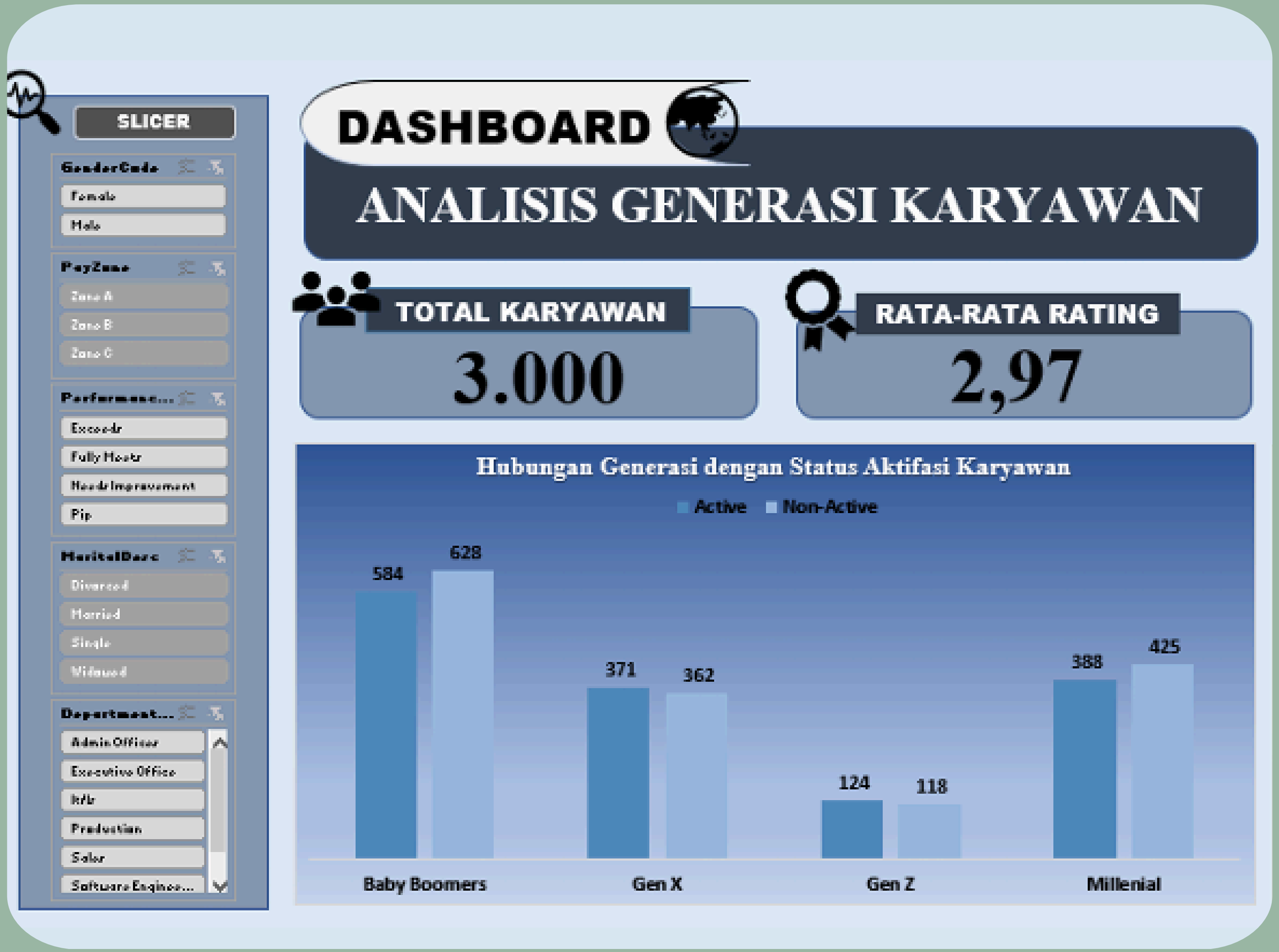
- Dibimbing Bootcamp (2026-sekarang)
- S1-S2 Peternakan (2020-2025)

Kontak

Email : yumitaaaaaa@gmail.com

Linkedin : linkedin.com/in/yumitaa16

PROJECT OVERVIEW



DASHBOARD EXCEL INTERACTIVE

- Perbedaan karakteristik generasi karyawan belum dianalisis secara menyeluruh, sehingga berpotensi menyebabkan ketidakefisienan dalam pengelolaan tenaga kerja
- Melakukan data handling, data preparation, membuat dashboard, mendapatkan insight

PROBLEM STATEMENT

"Bagaimana pengaruh kombinasi jenis material dan ukuran tas terhadap harga jual produk? Selain itu, material manakah yang memberikan nilai ekonomi tertinggi pada setiap kategori ukuran?"

DATA UNDERSTANDING

Sumber Data

Data kaggle berbentu csv, diimport ke jupyter

Jumlah baris dan kolom

Terdapat 12805 baris dan 10 kolom

IMPORT DATA

QUERY

```
import numpy as np
import pandas as pd

df = pd.read_csv("train_Student_Bag_Price_Dataset (yumi &lina).csv")
df_ori = df.copy()
df.head()
```

HASIL

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price
0	Nike	Nylon	Small	9.0	No	No	Tote	Black	9.333159	116.655111
1	Adidas	Polyester	Large	NaN	No	No	Backpack	Gray	26.258602	127.225657
2	Under Armour	Nylon	Medium	7.0	No	No	Backpack	Blue	15.049938	132.131249
3	Under Armour	Canvas	Large	8.0	Yes	No	Tote	Green	17.426660	NaN
4	Puma	NaN	Large	1.0	No	Yes	Backpack	Black	12.771516	86.334448

- Proses import data untuk memanggil data CSV
- Ditampilkan hasil data

DATA EXPLORATION

INFORMASI DATA

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12805 entries, 0 to 12804
Data columns (total 10 columns):
 #   Column               Non-Null Count  Dtype
---  ------               --------------  -----
 0   Brand                12152 non-null  object
 1   Material             12167 non-null  object
 2   Size                 12169 non-null  object
 3   Compartments         12161 non-null  float64
 4   Laptop Compartment   12156 non-null  object
 5   Waterproof           12140 non-null  object
 6   Style                12133 non-null  object
 7   Color                12144 non-null  object
 8   Weight Capacity (kg) 12154 non-null  float64
 9   Price                12181 non-null  float64
dtypes: float64(3), object(7)
memory usage: 1000.5+ KB
```

Terdapat 12.805 baris dan 10 kolom

CEK NILAI DATA

	Compartments	Weight Capacity (kg)	Price
count	12161.000000	12154.000000	12181.000000
mean	5.536058	17.523785	81.748538
std	2.859863	7.559455	39.271530
min	1.000000	5.000000	15.000000
25%	3.000000	11.030286	47.379405
50%	6.000000	17.591189	81.312176
75%	8.000000	24.167772	115.715245
max	10.000000	30.000000	150.000000

Jumlah Missing Value

Brand	653
Material	638
Size	636
Compartments	644
Laptop Compartment	649
Waterproof	665
Style	672
Color	661
Weight Capacity (kg)	651
Price	624
dtype: int64	

Persentase Missing Value

Brand	5.099570
Material	4.982429
Size	4.966810
Compartments	5.029285
Laptop Compartment	5.068333
Waterproof	5.193284
Style	5.247950
Color	5.162046
Weight Capacity (kg)	5.083952
Price	4.873096
dtype: float64	

Terdapat **6.493** data missing value

DUPLICATE

```
duplicate = df.duplicated().sum()
print(duplicate)
```

```
0
```

Tidak terdapat duplicate data dan typo pada data

```
Brand
Under Armour    2472
Puma            2434
Nike            2427
Jansport        2420
Adidas          2399
Name: count, dtype: int64
```

```
Material
Polyester    3099
Leather      3050
Canvas       3018
Nylon        3000
Name: count, dtype: int64
```

```
Size
Large     4083
Small     4046
Medium    4040
Name: count, dtype: int64
```

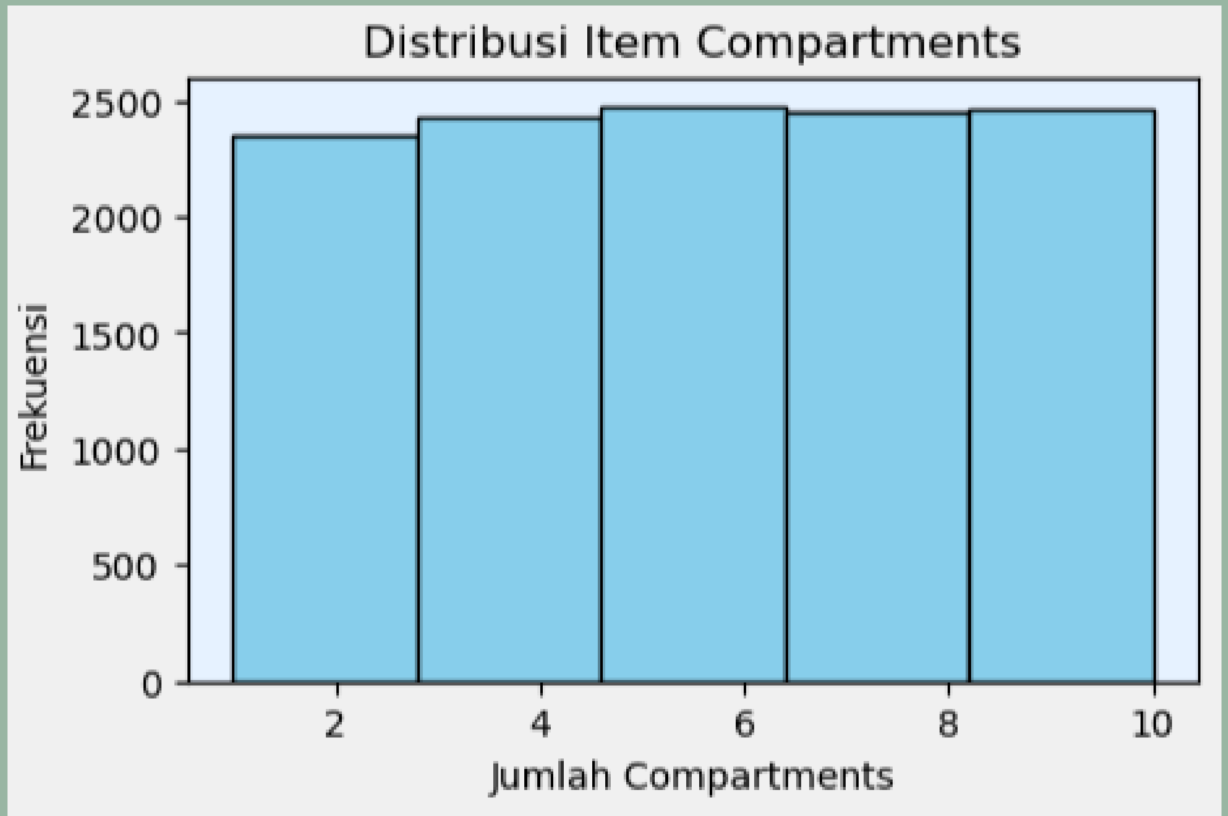
```
Laptop Compartment
No     6140
Yes    6016
Name: count, dtype: int64
```

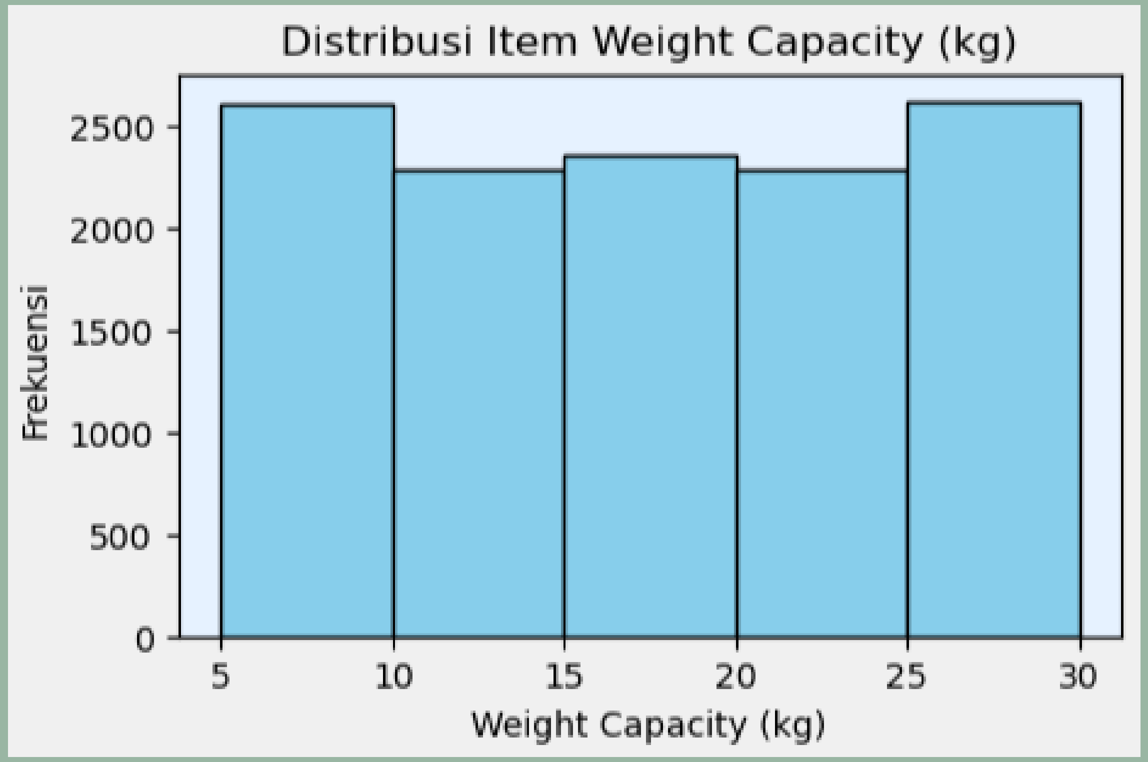
```
Waterproof
No     6167
Yes    5973
Name: count, dtype: int64
```

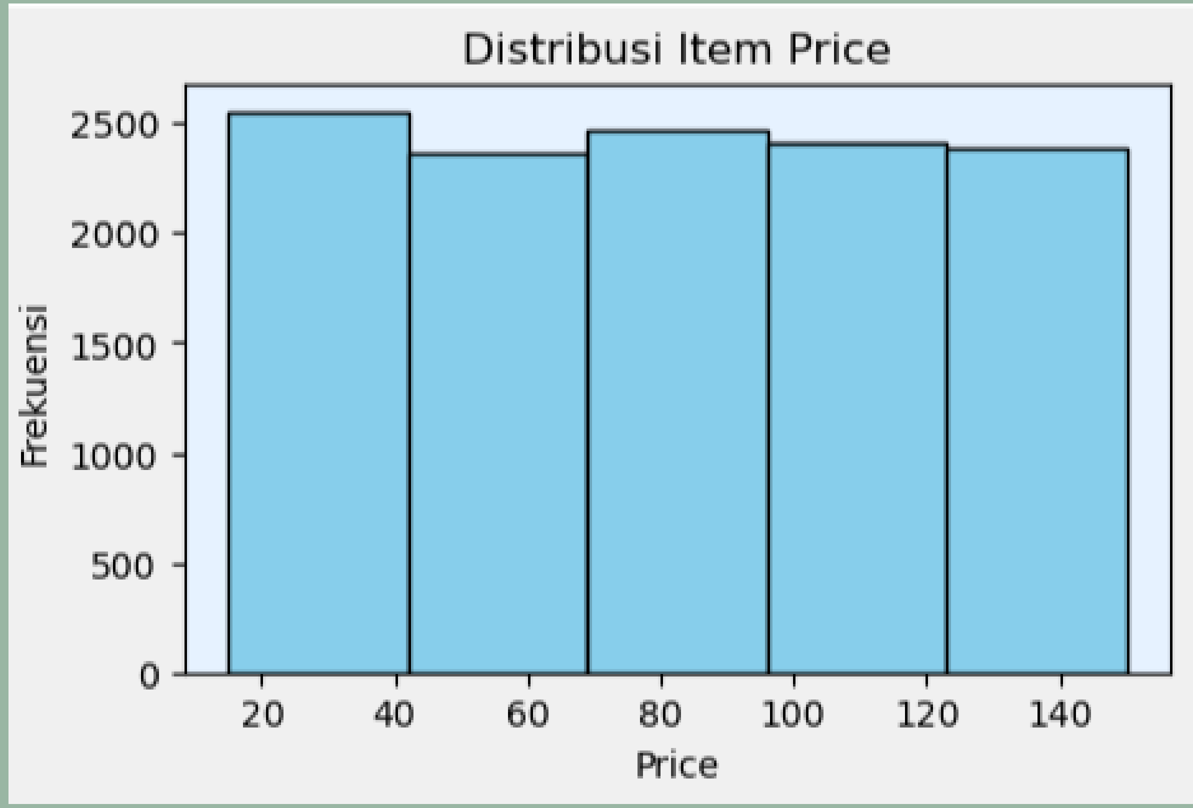
```
Style
Tote         4073
Messenger    4050
Backpack     4010
Name: count, dtype: int64
```

DISTRIBUSI DATA

Melihat penyebaran data yang nantinya dijadikan acuan untuk menangani missing value







DATA CLEANING

MENANGANI MISSING VALUE

- setelah diamati untuk misseng value data kategori menggunakan probabilitas karena penyebarang masing-masing nya sama, jadi dibagi rata untuk mengisi missing valuenya

SEBELUM

```
print(df_ori['Brand'].value_counts())
```

```
Brand
Under Armour    2472
Puma            2434
Nike            2427
Jansport        2420
Adidas          2399
Name: count, dtype: int64
```

SESUDAH

```
print(df_ori['Brand'].value_counts())
```

```
Brand
Under Armour    2600
Nike            2572
Puma            2559
Jansport        2551
Adidas          2523
Name: count, dtype: int64
```

MENANGANI MISSING VALUE

QUERY

```
np.random.seed(17)

total = 2472 + 2434 + 2427 + 2420 + 2399

mask = df_ori['Brand'].isin(['unknown']) | df_ori['Brand'].isnull()

df_ori.loc[mask, 'Brand'] = np.random.choice(
    ['Under Armour', 'Puma', 'Nike', 'Jansport', 'Adidas'],
    size = mask.sum(),
    p = [2472/total, 2434/total, 2427/total, 2420/total, 2399/total]
)
```

MENGHAPUS DUPLIKAT DATA

```
duplicate = df_ori.duplicated().sum()
print(duplicate)
```

```
0
```

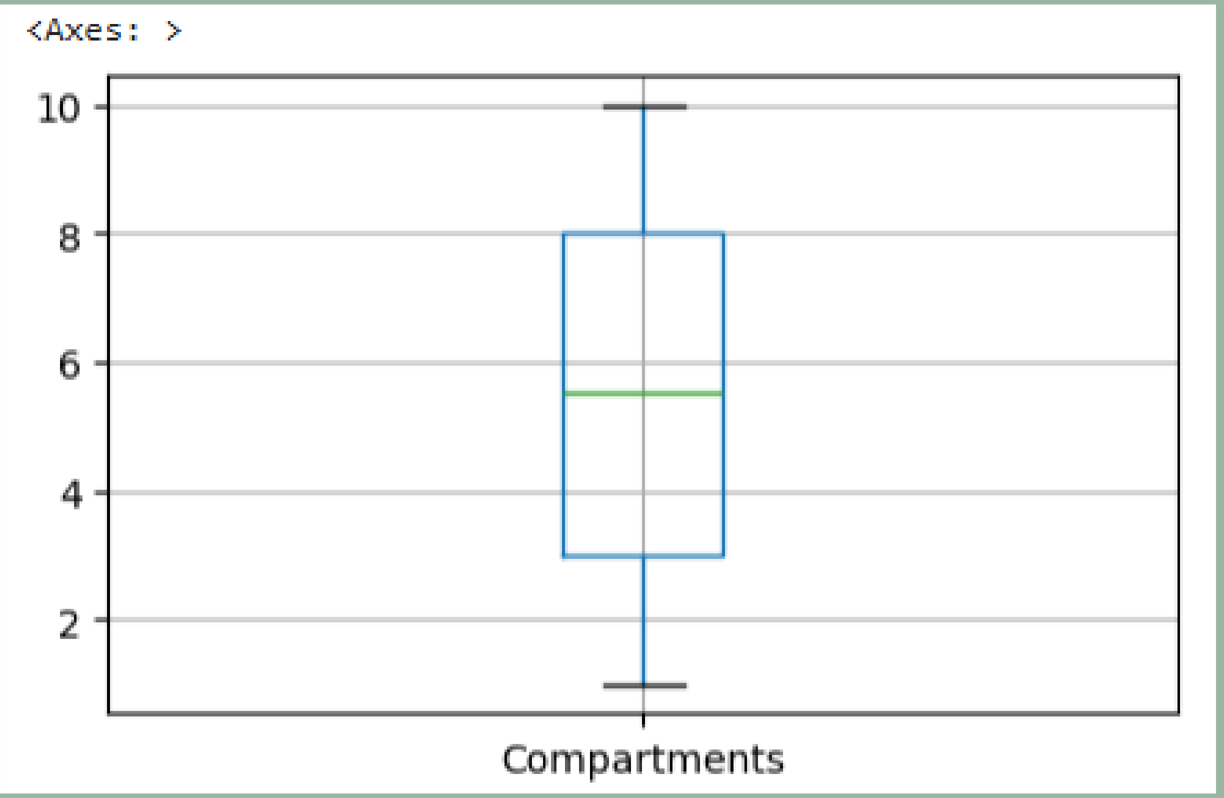
Setelah dicek ternyata tidak ada data yang duplikat sehingga tidak perlu diuji lanjut

MEMPERBAIKI INKONSISTENSI DATA

```
# Rapihin teks
df_ori2 = df_ori.copy()

df_ori2['Brand'] = df_ori2['Brand'].str.strip().str.title()
df_ori2['Material'] = df_ori2['Material'].str.strip().str.title()
df_ori2['Size'] = df_ori2['Size'].str.strip().str.title()
df_ori2['Laptop Compartment'] = df_ori2['Laptop Compartment'].str.strip().str.title()
df_ori2['Waterproof'] = df_ori2['Waterproof'].str.strip().str.title()
df_ori2['Style'] = df_ori2['Style'].str.strip().str.title()
df_ori2['Color'] = df_ori2['Color'].str.strip().str.title()
df_ori2.head()
```

Memperbaiki data kategori/object untuk spasi dan penggunaan huruf kapital diawal kata

MENGIDENTIFIKASI OUTLIER



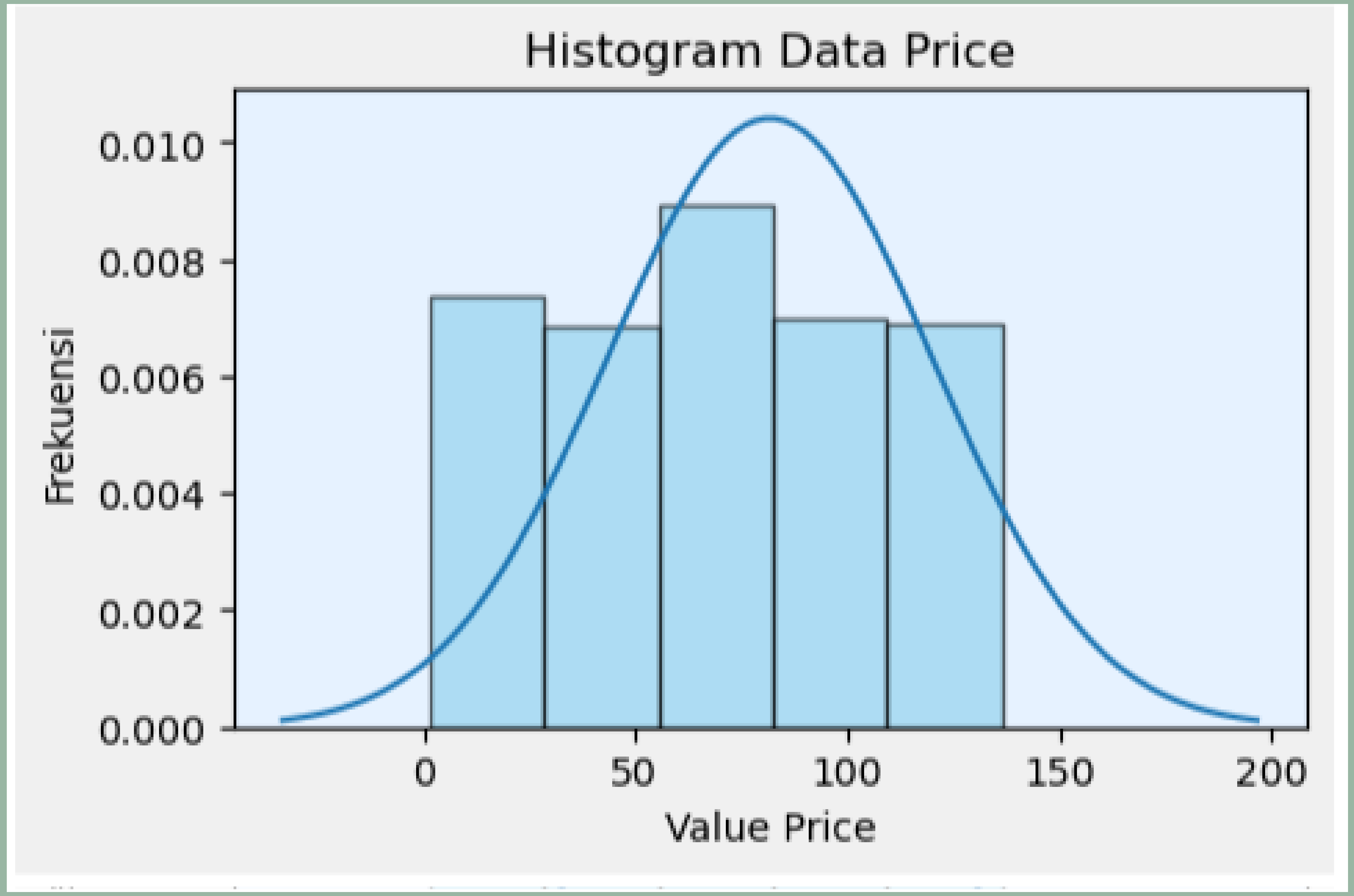

```
# deteksi outlier menggunakan IQR
df_ori3 = df_ori2.copy()

Q1=df_ori3['Compartments'].quantile(0.25)
Q3=df_ori3['Compartments'].quantile(0.75)
IQR=Q3-Q1

lower= Q1 - 1.5 * IQR
upper= Q3 + 1.5 * IQR

outlier_iqr = df_ori3[(df_ori3['Compartments'] < lower) | (df_ori3['Compartments'] > upper)]
print(outlier_iqr)
```

```
Empty DataFrame
Columns: [Brand, Material, Size, Compartments, Laptop Compartment, Waterproof, Style, Color, Weight Capacity (kg), Price]
Index: []
```

Dari identifikasi terhadap data numerik dapat diketahui bahwa tidak terdapat outlier.

DATA MANIPULATION

DISTRIBUSI DATA



- Setelah didapatkan histogram, lanjut uji normalitas
- jika distribusi tidak normal maka lanjut uji log transform, root transformtion, winsorize

Uji normalitas (uji Kolmogorov karena data >5000)

```
Statistik uji :  0.05479919436740105
p-value :  7.306857206083341e-34
Data tidak berdistribusi normal
```

Transformation (Log Transform)

```
Statistik uji :  0.11765978957881645
p-value :  6.566795643689488e-155
Data tidak berdistribusi normal
```

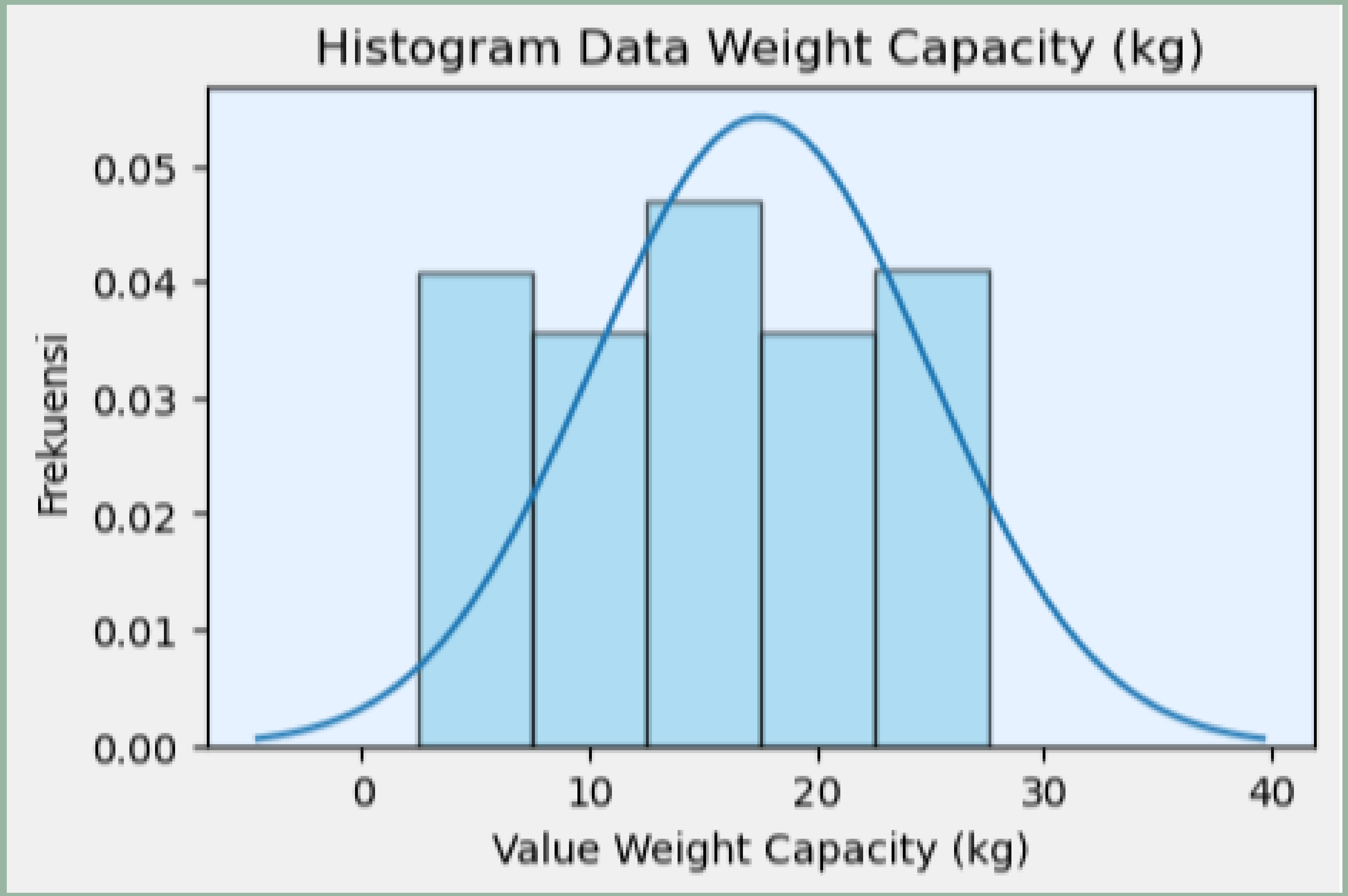
Transformation (Root Transformation)

```
Statistik uji :  0.07245086593990785
p-value :  6.768023618216784e-59
Data tidak berdistribusi normal
```

Transformasi (Winsourizing)

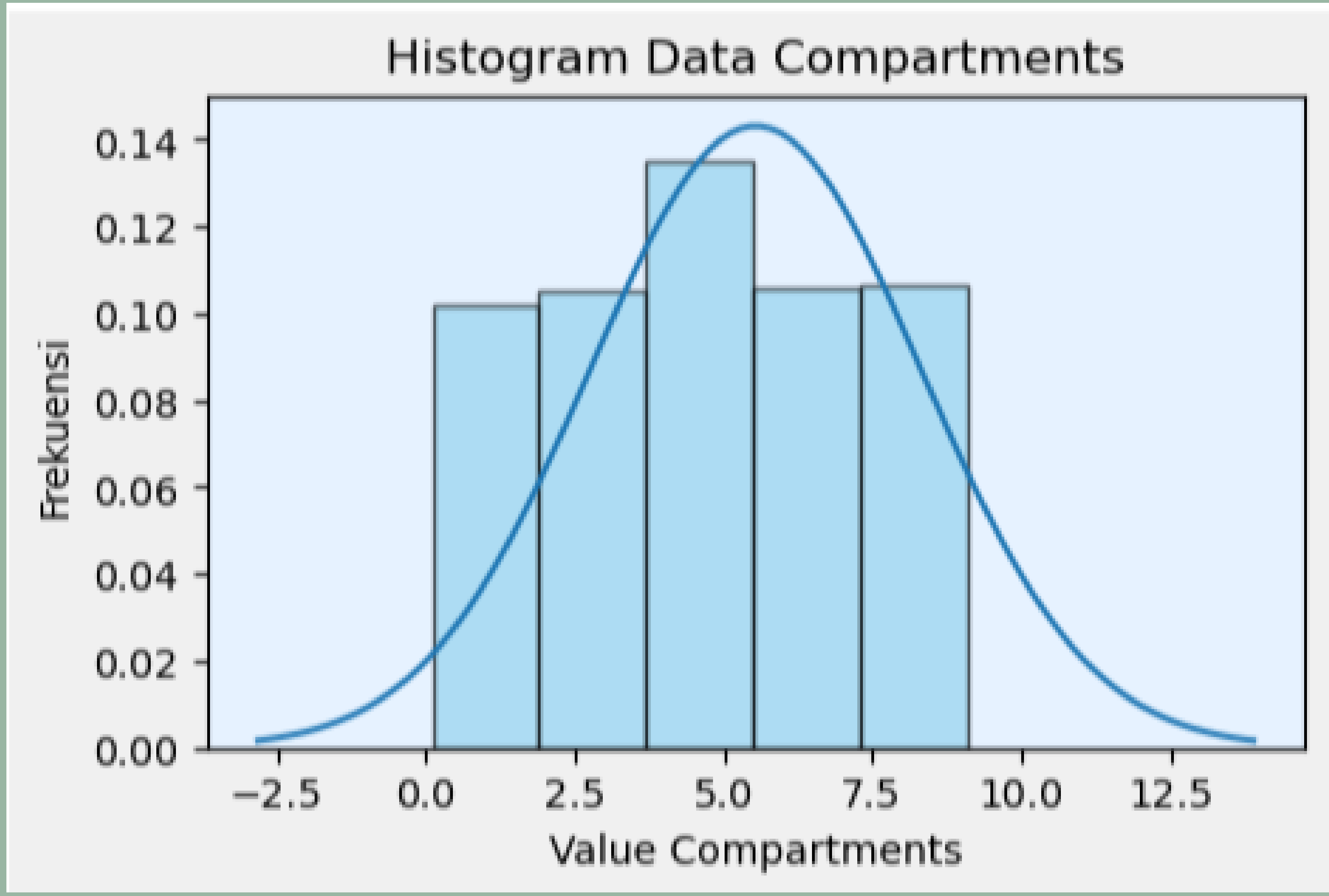
```
Statistik uji :  0.058432543858723596
p-value :  1.9071710088102165e-38
Data tidak berdistribusi normal
```

DISTRIBUSI DATA



Uji normalitas (uji Kolmogorov)

```
Statistik uji :  0.05412256320047315
p-value :  4.838453082497585e-33
Data tidak berdistribusi normal
```



Uji normalitas (uji Kolmogorov)

```
Statistik uji :  0.09899227264358201
p-value :  1.1043785078703025e-109
Data tidak berdistribusi normal
```

TYPE DATA CONVERSION

SEBELUM

```
Brand                  object
Material               object
Size                   object
Compartments          float64
Laptop Compartment     object
Waterproof             object
Style                  object
Color                  object
Weight Capacity (kg)  float64
Price                 float64
dtype: object
```

SESUDAH

```
Brand                  category
Material               category
Size                   category
Compartments              int64
Laptop Compartment     category
Waterproof             category
Style                  category
Color                  category
Weight Capacity (kg)    float64
Price                   float64
dtype: object
```

```
df_ori11 = df_ori10.copy()

# Menentukan kolok mana saja yang merupakan kategori
kolom_kategori = ['Brand', 'Material', 'Size', 'Laptop Compartment', 'Waterproof', 'Style', 'Color']

# Ubah semua sekaligus ke tipe category
df_ori11[kolom_kategori] = df_ori11[kolom_kategori].astype('category')
```

KATEGORI

NUMERIK

```
df_ori11['Compartments'] = df_ori11['Compartments'].astype(int)
df_ori11['Weight Capacity (kg)'] = df_ori11['Weight Capacity (kg)'].round(2)
df_ori11['Price'] = df_ori11['Price'].round(2)
```

FILTERING & SORTING

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price
5	Adidas	Canvas	Medium	10	Yes	No	Tote	Black	11.04	115.39
19	Nike	Canvas	Medium	9	Yes	No	Messenger	Green	29.05	90.90
36	Jansport	Canvas	Medium	5	Yes	No	Messenger	Red	23.50	56.43
39	Under Armour	Canvas	Medium	10	Yes	No	Tote	Black	15.77	114.19
42	Under Armour	Canvas	Medium	5	No	Yes	Backpack	Black	28.11	81.75

Melakukan filter terhadap material (Canvas) dan ukuran (medium)

Melakukan sorting data berdasarkan harga termahal

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price
3973	Nike	Nylon	Large	2	Yes	No	Tote	Red	26.00	150.0
6407	Nike	Canvas	Small	5	No	Yes	Messenger	Green	21.83	150.0
12071	Puma	Leather	Large	3	Yes	Yes	Backpack	Gray	22.02	150.0
8026	Jansport	Leather	Small	5	No	Yes	Messenger	Green	17.52	150.0
2150	Jansport	Canvas	Large	7	No	Yes	Tote	Gray	14.28	150.0

GRUPBYO

	Brand	rata_rata_harga	total_kapasitas	jumlah_produk
3	Puma	82.548163	44946.57	2559
2	Nike	81.675229	45358.03	2572
0	Adidas	81.564923	44339.66	2523
4	Under Armour	81.546196	45022.81	2600
1	Jansport	81.408593	44722.41	2551

- Data dilihat dari laporan berdasarkan brand
- Menampilkan hasil berupa rata-rata harga, total kapasitas, dan jumlab produk sesuai brandnya

FEATURE ENGINEERING

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price	Price per Compartments
0	Nike	Nylon	Small	9	No	No	Tote	Black	9.33	116.66	12.962222
1	Adidas	Polyester	Large	5	No	No	Backpack	Gray	26.26	127.23	25.446000
2	Under Armour	Nylon	Medium	7	No	No	Backpack	Blue	15.05	132.13	18.875714
3	Under Armour	Canvas	Large	8	Yes	No	Tote	Green	17.43	81.75	10.218750
4	Puma	Leather	Large	1	No	Yes	Backpack	Black	12.77	86.33	86.330000

- Membuat kolom baru mengenai harga per compartments yang menggunakan kolom price dan compartments
- Hasilnya akan memeunculkan kolom baru yaitu price per compartments

OUTLIER

- Setelah dilakukan uji outlier pada bagian cleaning data, dapat diketahui bahwa tidak ada data outlier pada variabel numerik, sehingga tidak perlu dilakukan tahap penganagan outlier menggunakan scaling.

ENCODING

Berdasarkan type data terdapat 7 kolom kategori dimana terbagi atas 2 jenis yaitu :

- Ordinal (ada urutan) : kolom 'Size'
- Nominal (tanpa urutan) : kolom 'Brand', 'Material', 'Laptop Compartment', 'Waterproof', 'Style', 'Color'

Ordinal Encoding

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price	size encoder
0	Nike	Nylon	Small	9.000000	No	No	Tote	Black	9.333159	116.655111	0.0
1	Adidas	Polyester	Large	5.536058	No	No	Backpack	Gray	26.258602	127.225657	2.0
2	Under Armour	Nylon	Medium	7.000000	No	No	Backpack	Blue	15.049938	132.131249	1.0
3	Under Armour	Canvas	Large	8.000000	Yes	No	Tote	Green	17.426660	81.748538	2.0
4	Puma	Leather	Large	1.000000	No	Yes	Backpack	Black	12.771516	86.334448	2.0

One-Hot Encoding

	Compartments	Weight Capacity (kg)	Price	Brand_Jansport	Brand_Nike	Brand_Puma	Brand_Under Armour	Material_Leather	Material_Nylon	Material_Polyester	...	Size_Small
0	9.000000	9.333159	116.655111	0	1	0	0	0	1	0	...	1
1	5.536058	26.258602	127.225657	0	0	0	0	0	0	1	...	0
2	7.000000	15.049938	132.131249	0	0	0	1	0	1	0	...	0
3	8.000000	17.426660	81.748538	0	0	0	1	0	0	0	...	0
4	1.000000	12.771516	86.334448	0	0	1	0	1	0	0	...	0

ANALISIS STATISTIK

TOPIK : PERBANDINGAN MATERIAL DAN SIZE TERHADAP HARGA TAS

Hipotesis

- H0 : Tidak ada perbedaan distribusi harga yang signifikan jika ditinjau dari jenis material dan ukuran tas.
- H1 : Terdapat perbedaan distribusi harga yang signifikan jika ditinjau dari jenis material dan ukuran tas.

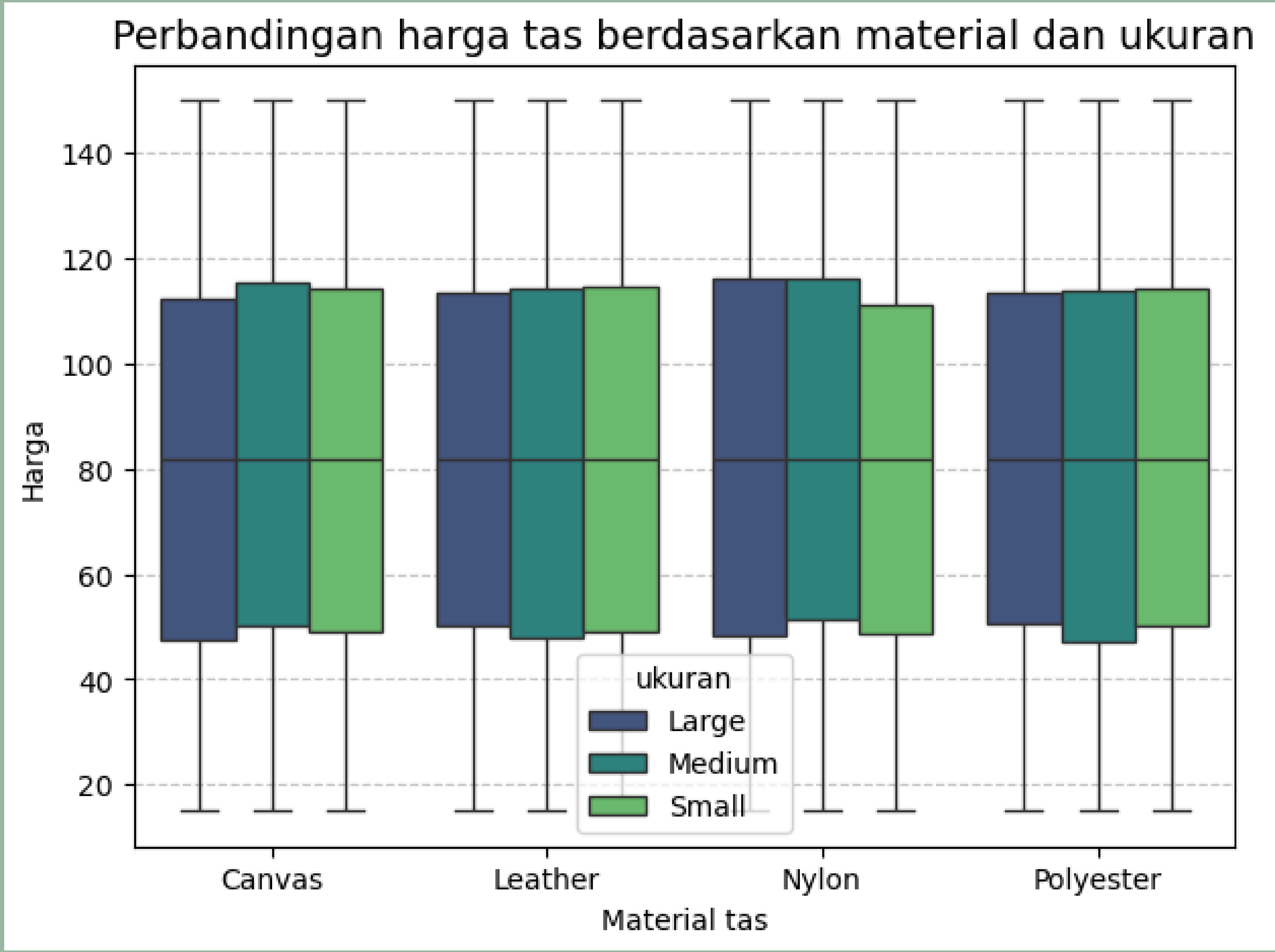
Brand	0
Material	0
Size	0
Compartments	0
Laptop Compartment	0
Waterproof	0
Style	0
Color	0
Weight Capacity (kg)	0
Price	0
dtype: int64	

Brand	category
Material	category
Size	category
Compartments	int64
Laptop Compartment	category
Waterproof	category
Style	category
Color	category
Weight Capacity (kg)	float64
Price	float64
dtype: object	

DATA CLEANING

VISUALISASI DATA :

GROUPED BOXPLOT (CODE)



INSIGHT

1. Dapat dilihat dari boxplot bahwa terdapat stabilitas harga antar material. Rata-rata harga tas cenderung stabil dan seragam di kisaran 80 dollar (median). Dari semua materian tidak ada harganya yang mendominasi secara ekstrem dibandingkan material lainnya.
2. Pengaruh ukuran terhadap harga juga tidak terlalu signifikan dilihat dari tas canvas dan leather ukuran medium menjadi populer sehingga harganya bersaing
3. Tas materila nylon dengan ukuran large dan medium memiliki harga yang hampir sama, sedangkan ukuran small sedikit lebih murah.
4. Material polyester harganya sangat kompetitif antara ukuran

ANALISIS PMF

INSIGHT :

1. Hampir semua hasil data dari analisis PMF menunjukkan nilai probabilitynya 0.08 sekitar 8% yang menandakan jumlah datanya seimbang.

2. Setiap kombinasi antara material dan size memiliki peluang kemunculan yang hampir sama, tidak ada jenis tas yang mendominasi populasi data secara signifikan.

3. Hasil PMF menunjukkan keseimbangan kategori yang sangat baik dan akan menguntungkan untuk proses pemodelan berikutnya karena model tidak akan bias terhadap salah satu material atau ukuran tertentu.

Tabel PMF Kombinasi Material & Size:

	Material	Size	Probability
0	Canvas	Large	0.084654
1	Canvas	Medium	0.078407
2	Canvas	Small	0.085357
3	Leather	Large	0.083327
4	Leather	Medium	0.085904
5	Leather	Small	0.081531
6	Nylon	Large	0.083405
7	Nylon	Medium	0.083014
8	Nylon	Small	0.080281
9	Polyester	Large	0.084420
10	Polyester	Medium	0.084654
11	Polyester	Small	0.085045

HASIL KRUSKAL-WALLIS TEST

```
Kruskal-Wallis Test
H-statistic: 0.7844115345961541
p-value: 0.8531918397915716

Kesimpulan:

Kesimpulan: Gagal Tolak H0 artinya tidak ada perbedaan distribusi harga yang signifikan jika ditinjau dari jenis material dan ukuran tas
```

OUTPUT FINAL DATASET

DATA SET SEBELUM DIANALISIS

	Brand	Material	Size	Compartments	Laptop Compartment	Waterproof	Style	Color	Weight Capacity (kg)	Price
0	Nike	Nylon	Small	9.0	No	No	Tote	Black	9.333159	116.655111
1	Adidas	Polyester	Large	NaN	No	No	Backpack	Gray	26.258602	127.225657
2	Under Armour	Nylon	Medium	7.0	No	No	Backpack	Blue	15.049938	132.131249
3	Under Armour	Canvas	Large	8.0	Yes	No	Tote	Green	17.426660	NaN
4	Puma	NaN	Large	1.0	No	Yes	Backpack	Black	12.771516	86.334448

DATA SET SESUDAH DIANALISIS

	Compartments	Weight Capacity (kg)	Price	Brand_Jansport	Brand_Nike	Brand_Puma	Brand_Under Armour	Material_Leather	Material_Nylon	Material_Polyester	...	Size_Small
0	9.000000	9.333159	116.655111	0	1	0	0	0	1	0	...	1
1	5.536058	26.258602	127.225657	0	0	0	0	0	0	1	...	0
2	7.000000	15.049938	132.131249	0	0	0	1	0	1	0	...	0
3	8.000000	17.426660	81.748538	0	0	0	1	0	0	0	...	0
4	1.000000	12.771516	86.334448	0	0	1	0	1	0	0	...	0

5 rows × 21 columns

DETAIL DATA SEBELUM DIANALISIS

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12805 entries, 0 to 12804
Data columns (total 10 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Brand                 12152 non-null  object
 1   Material              12167 non-null  object
 2   Size                  12169 non-null  object
 3   Compartments          12161 non-null  float64
 4   Laptop Compartment    12156 non-null  object
 5   Waterproof            12140 non-null  object
 6   Style                 12133 non-null  object
 7   Color                 12144 non-null  object
 8   Weight Capacity (kg)  12154 non-null  float64
 9   Price                 12181 non-null  float64
dtypes: float64(3), object(7)
memory usage: 1000.5+ KB
```

DETAIL DATA SESUDAH DIANALISIS

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 12805 entries, 0 to 12804
Data columns (total 21 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   Compartments            12805 non-null  float64
 1   Weight Capacity (kg)    12805 non-null  float64
 2   Price                   12805 non-null  float64
 3   Brand_Jansport          12805 non-null  int64
 4   Brand_Nike              12805 non-null  int64
 5   Brand_Puma              12805 non-null  int64
 6   Brand_Under Armour      12805 non-null  int64
 7   Material_Leather        12805 non-null  int64
 8   Material_Nylon          12805 non-null  int64
 9   Material_Polyester      12805 non-null  int64
 10  Size_Medium             12805 non-null  int64
 11  Size_Small              12805 non-null  int64
 12  Laptop Compartment_Yes  12805 non-null  int64
 13  Waterproof_Yes          12805 non-null  int64
 14  Style_Messenger         12805 non-null  int64
 15  Style_Tote              12805 non-null  int64
 16  Color_Blue              12805 non-null  int64
 17  Color_Gray              12805 non-null  int64
 18  Color_Green             12805 non-null  int64
 19  Color_Pink              12805 non-null  int64
 20  Color_Red               12805 non-null  int64
dtypes: float64(3), int64(18)
memory usage: 2.1 MB
```

RINGKASAN

1. Missing value
   - sebelum : 6493 (50,71%)
   - sesudah : 0

2. Duplicate data
   - sebelum : 0
   - sesudah : 0

3. Tipe Data
   - sebelum : object, float
   - sesudah : category, int, float

CONCLUSION

- Melalui eksplorasi data yang mendalam, saya menemukan bahwa meskipun dataset memiliki distribusi Price yang tidak normal, variabel Material dan Size terdistribusi secara sangat merata (berdasarkan hasil PMF sekitar 8%).

- Menariknya, uji Kruskal-Wallis dan Boxplot menunjukkan tidak adanya perbedaan harga yang ekstrem antar kategori material. Hal ini mengindikasikan bahwa produk memiliki strategi harga yang kompetitif dan stabil. Data ini sangat valid dan memiliki kualitas yang baik untuk dilanjutkan ke tahap pemodelan Machine Learning tanpa risiko bias yang signifikan.

THANK YOU

The Q&A session begins